

Shinsuke Maki

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About Me

I'm a Senior AI and Full-Stack Engineer with 20+ years designing, building and operating production software. I focus on LLM-powered applications, e.g., document-ingestion pipelines, embedding generation, hybrid retrieval, reranking, retrieval-augmented generation, structured outputs, and tool-using agent workflows. I build evaluation pipelines for measuring retrieval relevance, groundedness, latency, reliability, and model cost, with safety measures for prompt injection, data access, and output validation.

I have a full-stack background, which allows me to build complete AI products, not just model prototypes. I create python and fastAPI services, react and next.js user interfaces, postgresql data models, asynchronous processing workflows, and containerized cloud infrastructure. I have strong architectural judgment, ownership of production and technical leadership for projects requiring AI capabilities to operate securely, reliably and at scale.

Areas of Expertise

• Python	• TypeScript	• JavaScript	• React
• Next.js	• FastAPI	• Ruby on Rails	• Node.js
• PostgreSQL	• MySQL	• MongoDB	• Redis
• AWS	• Docker	• Linux	• Nginx
• Github Actions	• REST API	• LLM APIs	• RAG
• AI Agents	• Prompt Engineering	• Embeddings	• Vector Search
• Semantic Search	• Hybrid Search	• Reranking	• Vector Databases
• LangChain	• Structured Outputs	• Function Calling	• Document Processing
• LLM Evaluation	• Hallucination Testing	• Prompt Injection Protection	

Professional Experience

Senior LLM Systems & Full-Stack Engineer, Sakana AI Co., Ltd., Tokyo

Nov 2024 to Present

Architected and delivered production LLM applications across model integration, retrieval, agent orchestration, evaluation, security, frontend, backend, data, and cloud infrastructure. Led technical decisions required to operate AI capabilities reliably beyond the prototype stage.

Key Responsibilities

- Designed end-to-end architectures for production LLM and full-stack applications
- Selected models according to reasoning quality, latency, context limits, privacy, and operating cost
- Built multi-stage RAG pipelines using hybrid retrieval, query rewriting, metadata filtering, and reranking
- Developed tool-using agents that interacted with APIs, databases, and controlled business workflows
- Defined tool schemas, execution permissions, state transitions, retry limits, and human-approval points
- Implemented structured generation, output validation, fallback behavior, and model-routing strategies
- Created evaluation datasets and regression pipelines for retrieval and generation quality
- Monitored groundedness, task completion, failure patterns, latency, token consumption, and cost
- Applied safeguards against prompt injection, unauthorized retrieval, sensitive-data exposure, and unsafe tool execution
- Developed Python and FastAPI services, React and Next.js interfaces, and PostgreSQL data models
- Implemented asynchronous processing, Docker-based deployment, CI/CD, logging, and operational monitoring
- Led architecture planning, code reviews, technical prioritization, and production delivery

Selected Projects

- Agentic Workflow Automation Platform
- Built an LLM-based orchestration system that interpreted user requests, selected authorized tools, called internal APIs, and maintained state across multi-step workflows.
- Result: Automated complex operational processes while retaining validation, execution limits, auditability, and human approval for sensitive actions.
- Secure Enterprise RAG Platform
- Architected a multi-source knowledge system with document-level permissions, hybrid retrieval, reranking, citations, and evaluation pipelines.

Result: Delivered grounded answers from private organizational data while enforcing access restrictions throughout ingestion and retrieval.

- LLM Evaluation and Observability System

Implemented automated test suites and production monitoring for prompt versions, model configurations, retrieved context, generated outputs, latency, failures, and cost.

Result: Enabled evidence-based model and prompt changes, faster failure diagnosis, and safer production releases.

Technologies: Python, FastAPI, TypeScript, React, Next.js, LLM APIs, RAG, AI Agents, Function Calling, Structured Outputs, Embeddings, Vector Databases, Hybrid Retrieval, Reranking, PostgreSQL, Redis, Docker, AWS, CI/CD

Generative AI & RAG Engineer, BIX Tech Corp, Florida

Aug 2023 to Oct 2024

Designed and implemented retrieval-augmented generation systems that combined private document collections with large language models. Owned the pipeline from document ingestion and retrieval through prompt construction, generation, validation, and full-stack delivery.

Key Responsibilities

- Built document-ingestion pipelines for PDF, HTML, and structured business data
- Designed text-cleaning, chunking, metadata-extraction, and embedding workflows
- Stored and indexed document embeddings in vector databases
- Implemented semantic, keyword, and hybrid retrieval with metadata filtering
- Added reranking to improve the relevance of context supplied to language models
- Designed prompts that combined user queries with retrieved source material
- Implemented citations, structured outputs, schema validation, and response recovery
- Managed conversation history and model context-window limits
- Evaluated retrieval relevance, answer correctness, groundedness, and hallucination risk
- Optimized model selection, token usage, response latency, and API cost
- Integrated RAG services with React and Next.js applications

Selected Projects

- Enterprise Knowledge Assistant

Built a RAG application that ingested internal documents, retrieved relevant passages, and generated answers grounded in authorized source material.

Result: Provided a unified natural-language interface for accessing distributed organizational knowledge while preserving links to supporting evidence.

- Document Analysis and Structured Extraction Platform

Developed an LLM workflow for summarizing documents and extracting validated fields into predefined schemas.

Result: Transformed unstructured documents into consistent, searchable data for downstream business workflows.

- RAG Evaluation Framework

Created test datasets and automated evaluation workflows for retrieval relevance, groundedness, answer completeness, latency, and cost.

Result: Made prompt, model, and retrieval changes measurable and reduced the risk of undetected quality regressions.

Technologies: Python, FastAPI, OpenAI API, LangChain, RAG, Embeddings, Vector Databases, Hybrid Search, Reranking, PostgreSQL, React, Next.js, TypeScript, Docker, AWS

NLP & AI Application Engineer, BIX Tech Corp, Florida

Apr 2021 to Jul 2023

Developed AI-enabled application features for processing, searching, classifying, and summarizing unstructured text. Combined Python-based model services with existing web applications and operational workflows.

Key Responsibilities

- Built Python services for text classification, summarization, and information extraction
- Integrated hosted NLP and language-model APIs with existing business applications
- Developed pipelines for extracting and normalizing text from documents and web content
- Divided documents into searchable segments and generated vector embeddings
- Implemented semantic search based on similarity between queries and document embeddings
- Exposed model and search capabilities through FastAPI endpoints
- Added validation, retry, timeout, and error-handling controls around external model APIs
- Measured response quality, processing latency, token consumption, and API cost
- Integrated AI services with frontend applications and internal business tools

Selected Projects

- Semantic Document Search System

Built a document-processing pipeline that extracted text, generated embeddings, and retrieved relevant content using vector similarity.

Result: Enabled users to locate contextually relevant information without relying solely on exact keyword matches.

- Automated Text Classification Service

Developed a Python service that classified incoming documents and messages by topic, intent, or workflow category.

Result: Reduced manual sorting and provided consistent routing of unstructured content to downstream processes.

- Language-Model API Integration

Integrated external language-model APIs to generate summaries and extract structured fields from business documents.

Result: Converted lengthy unstructured content into concise summaries and application-ready data.

Technologies: Python, FastAPI, NLP APIs, Language-Model APIs, Embeddings, Vector Search, PostgreSQL, React, TypeScript, Docker, AWS

Senior Full-Stack & Data Systems Engineer, BIX Tech Corp, Florida

May 2018 to Mar 2021

Led the design and implementation of production web applications across frontend, backend, database, and infrastructure layers. Took ownership of system architecture, data-processing workflows, performance, reliability, and technical delivery.

Key Responsibilities

- Designed end-to-end architectures for customer-facing and administrative applications
- Developed frontend interfaces using React, Vue.js, JavaScript, and TypeScript
- Built backend services, REST APIs, and background-processing workflows
- Designed data-ingestion, validation, transformation, and reporting processes
- Optimized database schemas, indexes, queries, caching, and asynchronous jobs
- Implemented authentication, role-based authorization, and security controls
- Containerized application services and standardized development environments with Docker
- Established CI/CD workflows for automated testing and deployment
- Monitored production systems and resolved performance and reliability issues
- Reviewed code, planned technical implementation, and supported other engineers

Selected Projects

- Online Marketplace Modernization

Modernized a production marketplace across customer interfaces, administrative functions, backend services, and database workflows. Refactored legacy components and introduced clearer service boundaries for order, payment, and account operations.

Result: Improved system maintainability and created a more reliable foundation for introducing new business features.

- Transaction Data and Reporting Platform

Developed data-processing workflows and administrative reporting functions for orders, payments, refunds, and account activity. Implemented filtering, aggregation, export, and scheduled processing for operational data.

Result: Centralized business-critical information and enabled operations teams to investigate transactions and generate reports more efficiently.

- Containerized Development and Deployment Workflow

Standardized application services, databases, and supporting processes using Docker and automated build and deployment pipelines.

Result: Reduced environment inconsistencies, improved release repeatability, and made application setup more reliable for the engineering team.

Technologies: TypeScript, JavaScript, React, Vue.js, Ruby, Ruby on Rails, REST APIs, PostgreSQL, MySQL, Redis, Sidekiq, Docker, AWS, Linux, Nginx, GitHub Actions

Backend Engineer, Kistem Co., Ltd., Nara-shi

Sept 2013 to Apr 2018

Designed and maintained backend services for database-driven web applications. Progressed from feature implementation to ownership of business logic, data integrity, external integrations, and production reliability.

Key Responsibilities

- Developed backend applications and REST APIs using Ruby on Rails
- Translated business requirements into application services, data models, and processing workflows
- Designed relational database schemas, constraints, indexes, and migrations
- Implemented authentication, authorization, validation, and transaction processing
- Integrated payment, notification, and other third-party APIs
- Developed background jobs for asynchronous and scheduled processing
- Optimized database queries and investigated application-performance bottlenecks

- Created automated tests for business-critical backend functionality
- Diagnosed production incidents and implemented permanent corrective changes
- Participated in code reviews, releases, and ongoing system maintenance

Selected Projects

- Online Marketplace and Transaction Management System

Developed backend functionality for user accounts, product listings, orders, payments, cancellations, and administrative operations.

Result: Established a centralized transaction workflow and improved the consistency of order and payment records across customer and administrative functions.

- External Payment Service Integration

Implemented server-side payment authorization, transaction-status handling, callback processing, and failure-recovery workflows.

Result: Enabled secure online payments and improved transaction reliability by handling incomplete, failed, and duplicated requests consistently.

- Background Processing and Notification System

Built asynchronous jobs for email delivery, transaction updates, scheduled operations, and other long-running processes.

Result: Moved time-consuming work outside synchronous web requests, improving application responsiveness and operational stability.

Technologies: Ruby, Ruby on Rails, JavaScript, REST APIs, PostgreSQL, MySQL, Redis, Linux, Nginx, Git

Frontend Engineer, Sojin Co., Ltd., Nara-shi

Jun 2009 to Aug 2013

Developed production websites and browser-based interfaces for corporate clients. Transformed visual designs and business requirements into responsive, maintainable frontend implementations while collaborating with designers and backend engineers.

Key Responsibilities

- Converted design mockups into standards-compliant HTML and CSS
- Developed interactive user-interface components using JavaScript and jQuery
- Integrated frontend pages with backend systems and content management platforms
- Implemented form validation, navigation, dynamic content, and asynchronous data updates
- Ensured consistent behavior across major browsers and device sizes
- Improved page structure, loading performance, usability, and maintainability
- Diagnosed and resolved layout, styling, and client-side scripting defects
- Supported production releases and maintained existing client websites

Selected Projects

- Corporate Website Development and Redesign

Implemented multi-page corporate websites from designer-provided layouts, including reusable page structures, navigation components, inquiry forms, and content sections.

Result: Delivered consistent, production-ready websites that accurately reflected client designs and simplified subsequent content updates.

- E-commerce Frontend Implementation

Developed product listing, product detail, shopping-cart, registration, and checkout interfaces and integrated them with server-side order-processing functions.

Result: Created a consistent purchasing workflow and improved the usability of customer-facing commerce functions.

- Content Management System Templates

Built reusable frontend templates and components for database-driven corporate websites and publishing systems.

Result: Standardized page presentation across websites and enabled content updates without requiring repeated frontend implementation.

Technologies: HTML, CSS, JavaScript, jQuery, AJAX, PHP, MySQL

Web Application Developer, Nara-shi

Apr 2005 to May 2009

Independent web application developer, completed projects for Japanese and international clients through direct contracts, referrals and online collaboration.

Key Responsibilities

- Developed database driven web applications based on the client requirements
- Developed UI, server-based logic and database operations.
- Converted business requirements into technical specifications and workflows

- Designed relational database schemas for customers, reservations, products and orders
- Integrated front end components with server- side logic.
- Developed application defects and maintained deployed systems

Selected Projects

- Online Booking and Client Management System

Developed a web application for reservation booking, availability checking and maintaining customer and booking records with the help of an administrative interface.

Result: Centralized reservation management, reduced manual record-keeping, and improved the accuracy of booking information.

- Order Management & E-commerce Platform

Implemented product listings, shopping-cart functionality, customer registration, order processing, inventory updates, and administrative order management.

Result: Established an integrated workflow from online purchase through order fulfillment and simplified day-to-day order administration.

- Corporate Content Management System

Built a database-driven content management system that allowed authorized staff to publish and update pages, announcements, and company information.

Result: Enabled clients to manage routine website updates independently and publish new content more efficiently.

Technologies: HTML, CSS, JavaScript, PHP, MySQL, Apache, Linux

Education

NIIGATA UNIVERSITY, Niigata

Apr 2001 to Mar 2005

Studied Software Engineering, with a focus on programming, algorithms, database systems, software design, and the fundamentals of web application development.

Languages

Japanese, English, Chinese

Hobbies

Football, Badminton